

```
(kali㉿kali)-[~/vaje16]
$ echo "API_KEY=super-secret-key-123
DB_PASSWORD=VeryStrongPassword" > secrets.env
```

```
(kali㉿kali)-[~/vaje16]
$ cat secrets.env
API_KEY=super-secret-key-123
DB_PASSWORD=VeryStrongPassword
```

```
(kali㉿kali)-[~/vaje16]
$ gpg -c secrets.env
```

```
(kali㉿kali)-[~/vaje16]
$ cat secrets.env.gpg
```

```
♦
Y++bX++X++
♦BM++++#D++$XOCaC:++J++[++L+k          ♦♦♦♦VA♦♦♦8♦♦6t♦25v♦n9]ob♦w♦♦♦♦♦♦♦♦♦♦j♦♦K/♦_Ig♦♦♦[♦♦♦♦}♦j1♦♦w♦♦yYfp♦\♦♦♦
```

```
(kali㉿kali)-[~/vaje16]
$ cat secrets.env
API_KEY=super-secret-key-123
DB_PASSWORD=VeryStrongPassword
```

```
(kali㉿kali)-[~/vaje16]
$ rm secrets.env
```

```
(kali㉿kali)-[~/vaje16]
$ cat secrets.env.gpg
```

```
♦
Y++bX++X++
♦BM++++#D++$XOCaC:++J++[++L+k          ♦♦♦♦VA♦♦♦8♦♦6t♦25v♦n9]ob♦w♦♦♦♦♦♦♦♦♦♦j♦♦K/♦_Ig♦♦♦[♦♦♦♦}♦j1♦♦w♦♦yYfp♦\♦♦♦
```

```
(kali㉿kali)-[~/vaje16]
$ ls
secrets.env.gpg
```

```
(kali㉿kali)-[~/vaje16]
$ gpg secrets.env.gpg
gpg: WARNING: no command supplied. Trying to guess what you mean ...
gpg: AES256.CFB encrypted data
gpg: encrypted with 1 passphrase
```

```
(kali㉿kali)-[~/vaje16]
$ ls
secrets.env  secrets.env.gpg
```

```
(kali㉿kali)-[~/vaje16]
$ cat secrets.env
API_KEY=super-secret-key-123
DB_PASSWORD=VeryStrongPassword
```

```
(kali㉿kali)-[~/vaje16]
$
```

```

(kali㉿kali)-[~/vaje16]
$ ls
secrets.env  secrets.env.gpg

(kali㉿kali)-[~/vaje16]
$ cat secrets.env
API_KEY=super-secret-key-123
DB_PASSWORD=VeryStrongPassword

(kali㉿kali)-[~/vaje16]
$ gpg -d secrets.env.gpg
gpg: AES256.CFB encrypted data
gpg: encrypted with 1 passphrase
API_KEY=super-secret-key-123
DB_PASSWORD=VeryStrongPassword

(kali㉿kali)-[~/vaje16]
$ gpg -d secrets.env.gpg | grep API_KEY
gpg: AES256.CFB encrypted data
gpg: encrypted with 1 passphrase
API_KEY=super-secret-key-123

(kali㉿kali)-[~/vaje16]
$ gpg --full-generate-key
gpg (GnuPG) 2.4.8; Copyright (C) 2025 g10 Code GmbH
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

Please select what kind of key you want:
  (1) RSA and RSA
  (2) DSA and Elgamal
  (3) DSA (sign only)
  (4) RSA (sign only)
  (9) ECC (sign and encrypt) *default*
  (10) ECC (sign only)
  (14) Existing key from card
Your selection? 1
RSA keys may be between 1024 and 4096 bits long.
What keysize do you want? (3072) 4096
Requested keysize is 4096 bits
Please specify how long the key should be valid.
    0 = key does not expire
    <n>  = key expires in n days
    <n>w  = key expires in n weeks
    <n>m  = key expires in n months
    <n>y  = key expires in n years
Key is valid for? (0) 3m
Key expires at Tue 17 Mar 2026 01:18:30 PM EDT
Is this correct? (y/N) y

GnuPG needs to construct a user ID to identify your key.

Real name: App FIS Vaje

```

We need to generate a lot of random bytes. It is a good idea to perform some other action (type on the keyboard, move the mouse, utilize the disks) during the prime generation; this gives the random number generator a better chance to gain enough entropy.
gpg: revocation certificate stored as '/home/kali/.gnupg/openpgp-revocs.d/1ABA8D16235DB6961187C8FF3646FE511A864396.rev'
public and secret key created and signed.

```
pub  rsa4096 2025-12-17 [SC] [expires: 2026-03-17]
     1ABA8D16235DB6961187C8FF3646FE511A864396
uid          App FIS Vaje (FIS KV Vaje) <app@fis.vaje>
sub  rsa4096 2025-12-17 [E] [expires: 2026-03-17]
```

(kali@kali)-[~/vaje16]

\$ gpg --list-keys

gpg: checking the trustdb
gpg: marginals needed: 3 completes needed: 1 trust model: pgp
gpg: depth: 0 valid: 5 signed: 0 trust: 0-, 0q, 0n, 0m, 0f, 5u
gpg: next trustdb check due at 2026-03-17
/home/kali/.gnupg/pubring.kbx

```
pub  rsa4096 2025-12-17 [SC] [expires: 2026-12-17]
     9EEDFB1542D4D8D0916EB1A18DF495E7B56EC5CD
uid          [ultimate] Zans Test (KV-Vaje) <zans@kv.test>
sub  rsa4096 2025-12-17 [E] [expires: 2026-12-17]

pub  rsa4096 2025-12-17 [SC] [expires: 2026-12-17]
     2C88FDF21C89B21CD491009E9AA2BDF0968E3AB
uid          [ultimate] Zans2 Test (Vaje2) <zankv2@kv.test>
sub  rsa4096 2025-12-17 [E] [expires: 2026-12-17]

pub  rsa4096 2025-12-17 [SC] [expires: 2026-12-17]
     C7C9F95E071519793B33CF32BD124003A7D6E5CE
uid          [ultimate] Alice (AliceVaje) <alice@example.com>
sub  rsa4096 2025-12-17 [E] [expires: 2026-12-17]

pub  rsa4096 2025-12-17 [SC] [expires: 2026-12-17]
     B2202CAE05E9113953D5CF552A4F5E6308C00E1C
uid          [ultimate] Bob (BobVaje) <bob@example.com>
sub  rsa4096 2025-12-17 [E] [expires: 2026-12-17]

pub  rsa4096 2025-12-17 [SC] [expires: 2026-12-17]
     01FCDA6A9324E4E94EEB86D6A04B16DB60A9D046
uid          [ unknown] Mallory (MalloryVaje) <mallory@example.com>
sub  rsa4096 2025-12-17 [E] [expires: 2026-12-17]

pub  rsa4096 2025-12-17 [SC] [expires: 2026-03-17]
     1ABA8D16235DB6961187C8FF3646FE511A864396
uid          [ultimate] App FIS Vaje (FIS KV Vaje) <app@fis.vaje>
sub  rsa4096 2025-12-17 [E] [expires: 2026-03-17]
```

(kali@kali)-[~/vaje16]

\$

```

***c
Y1$C++j+3++p++A++;u{+++/B+t%+9++++J++S++P+H *[*0^w+++\\+++86+++6+++++LY3D+|+@++++g++>+++W+++l+++b9. V++?;+;+cSuu+?l+++6+++<
+)}+++dIZM+y++
'!+G+C++(!!+X++H+E+++++++f+++c++ p+7+3V+0C+teV+|d}+m+R+1-++HaD+B+f+3+0s++++u`+T
+D++{8+`++$|+ E6+z+++++
+~+ste+.+@+Afy*x0^
++%++UU+b+++.++s++.+F++F+++'[(*|+^+S+++@+/+++W!{+{d:+G++m++E_++o{+++++ +

(kali@kali)-[~/vaje16]
$ gpg secrets.env.gpg
gpg: WARNING: no command supplied. Trying to guess what you mean ...
gpg: AES256.CFB encrypted data
gpg: encrypted with 1 passphrase
File 'secrets.env' exists. Overwrite? (y/N) y

(kali@kali)-[~/vaje16]
$ cat secrets.env
API_KEY=super-secret-key-123
DB_PASSWORD=VeryStrongPassword

(kali@kali)-[~/vaje16]
$ gpg --encrypt --recipient app@fis.vaje secrets.env > secrets.env.asy
File 'secrets.env.gpg' exists. Overwrite? (y/N) y

(kali@kali)-[~/vaje16]
$ ls
secrets.env secrets.env.asy secrets.env.gpg

(kali@kali)-[~/vaje16]
$ diff secrets.env.asy secrets.env.gpg
Binary files secrets.env.asy and secrets.env.gpg differ

(kali@kali)-[~/vaje16]
$ cat secrets.env.asy

(kali@kali)-[~/vaje16]
$ cat secrets.env.gpg
+
+Z++S++w+E+++++W++++PA+++
Z.+
+++0++5[l++
+--+--+--+--+V+--+MB"++a+:+++r7+
+|+W}++-z?4,ex\+@ 8+gg*(++++Es+6[+9++B+++^8++U+ih+++u+?++d+W++b+W_@a++
+
p9e+++++,N+++++++teT++ss++Lrr!|+=+++++*+i+^+#+t+++#x"+++++Q;(+*l+
"3~+
+@++V-i+dg++++
t+v+++++M `++++@%+#++++++r+++++ly++++-D++++a++..+ej+h+++g+E+_ )WC++nz{,}+*W++++a+++++ck+n++
+x+4++++L+wX+(1+++5++++(++++3+p++G+++!+W@Z++{<%c+qf+7+9$++++cXk+}*o|+Y_++++++Y8+)*U+r+k++++
+@Hcd
8]*[T+fK,-+a/++++7

(kali@kali)-[~/vaje16]
$

(kali@kali)-[~/vaje16]
$ gpg --encrypt --recipient app@fis.vaje secrets.env
File 'secrets.env.gpg' exists. Overwrite? (y/N) y

(kali@kali)-[~/vaje16]
$ cat secrets.env.gpg
+
+Z++S++w+\\;Zx+++++0l+++#+pP+MV5?*+@++++++/z++Cr""+++++;+ +a+~+@Ou+*(+iBt+0+++++,=+++g)*++N `++C+0B++i+++3++++t+J]M++l+w++9+
u4]^+h+z+++++Vae+N++
W++++W+_+++ +Ks+Pv+E
+++++fp+Oj\.@+O+u+E+YjH;
+ )+F9+++++K++..+++++ [Q: _++n/++=++p+g5++z++@9+f+Z+r+++@W+u++iyf+c
+++++j++

U++++eL+gF++++X+h+C++++4++l+51++LQ+<Ul+Lj.+++
'+++++{m+1+YM+`-+++++~+yP+K_+++wJ+++++Z++pX++c+++m+\\+^++Q+++++3+8+F++++Y+
+J}++
+!,++
+m+a+J+u+++ ,3>+++++èK

(kali@kali)-[~/vaje16]
$ cat secrets.env
API_KEY=super-secret-key-123
DB_PASSWORD=VeryStrongPassword

(kali@kali)-[~/vaje16]
$ source secrets.env

(kali@kali)-[~/vaje16]
$ cat $DB_PASSWORD
cat: VeryStrongPassword: No such file or directory

(kali@kali)-[~/vaje16]
$ echo $DB_PASSWORD
VeryStrongPassword

(kali@kali)-[~/vaje16]
$ echo $API_KEY
super-secret-key-123

(kali@kali)-[~/vaje16]
$ unset API_KEY

(kali@kali)-[~/vaje16]
$ echo $API_KEY

(kali@kali)-[~/vaje16]
$

```

Razmislek:

1. Skrivnosti ne sodijo v izvirno kodo ker je koda shranjena velikokrat v git repozitorijih, preko katerih lahko dostopa več ljudi. Prav tako lahko vsak ki dostopa do te kode, vidi to skrivnost.
2. Simetrično šifriranje: za šifriranje in dešifriranje se uporabi isto geslo. Asimetrično šifriranje: uporabi se javni in zasebni ključ.
3. Šifriranih podatkov ne moremo več dešifrirati.
4. Uporabi se lahko centralizirano upravljanje skrivnosti z različnimi orodji. Dostop pa omejimo na vloge uporabnikov.